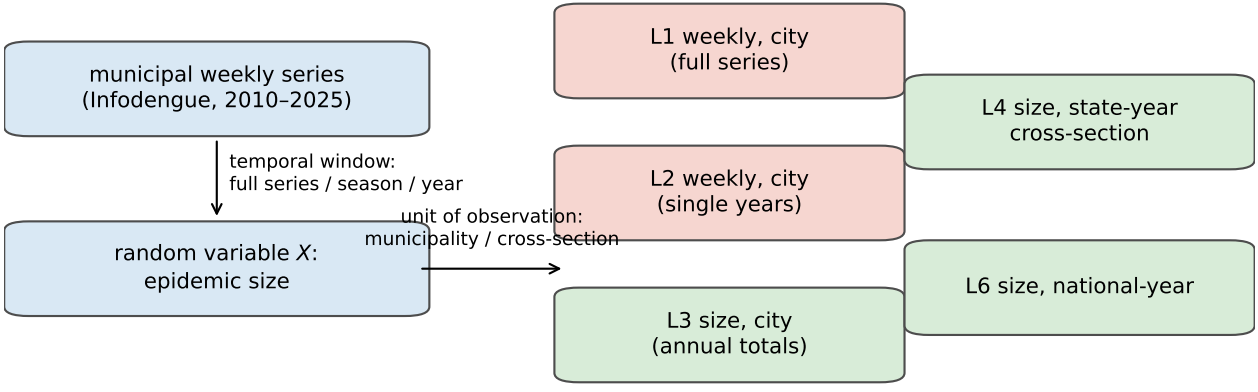

green: $x_{\min}$ above the bulk (the fit describes a tail) red: $x_{\min} \leq 5$ (the fit describes the data)